

# EPOC: An Efficient Pulse Generation Framework with Advanced Synthesis for Quantum Circuits

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**Abstract**—In this work, we aim to address the computational overhead challenge in quantum optimal control while reducing circuit latency. We propose a novel approach combining ZX-Calculus, circuit partitioning, and circuit synthesis for pulse generation. By implementing finer granularity in pulse generation and exploring equivalent circuit representations, we achieve increased parallelism and decreased latency. Our method demonstrates a 31.74% reduction in latency compared to previous work and a 76.80% reduction compared to gate-based pulse generation methods, while minimizing computational overhead.

**Index Terms**—quantum optimal control, quantum compilation, quantum circuit optimization

## I. INTRODUCTION

Quantum computing has emerged as a promising paradigm with the potential to outperform classical computers in various domains [1], including chemistry [2], machine learning [3], and physical simulations [4]. However, the current Noisy Intermediate-Scale Quantum (NISQ) era [5] presents significant challenges, such as limited qubit count, short coherence times, and high error rates [6]. To exploit the power of NISQ devices effectively, it is crucial to develop efficient quantum circuit optimization techniques that mitigate the impact of noise and errors [7]. The coherence time determines the duration and depth of quantum circuits that can be successfully executed on real quantum hardware. The term “quantum volume” has been proposed to quantitatively describe this property, which takes into account both the number of qubits and the depth of the quantum circuit that can be reliably executed on a given quantum device [8].

A typical quantum program compilation workflow starts with implementing quantum algorithms as quantum circuits. As shown in Figure 1, these circuits undergo compilation passes to become compatible with the hardware topology, including decomposition into basis gates and mapping to the target architecture [9]. Once expressed in basis gates, the gates are converted into physical signals applied to the qubits [10]. For superconducting quantum computers, these physical signals are modulated microwave pulses sent to transmons [11]. The modulated pulses are generated with an envelope that determines their function and a carrier signal.

In the final step of the compilation process, where basis gates are translated to pulses, quantum optimal control (QOC) [12] is used to find optimal control pulses that implement desired unitary operations, minimizing noise and errors [13]. However, QOC has significant computational overhead, limiting its scalability [14]. Previous works have proposed methods to accelerate QOC, such as utilizing GPUs, pre-compilation techniques, and improved circuit libraries [15], [16], [17]. These pulse generation frameworks focus on generating pulses from unitary matrices directly derived from quantum circuits without exploring equivalent representations, thus overlooking optimization opportunities by not adopting coarse-grained methods.

By coarse-grained methods, we refer to existing frameworks such as AccQOC [18] and PAQOC [19], which perform circuit partitioning at the gate level. These methods first group the quantum gates in the circuit and calculate the corresponding unitary matrices for each

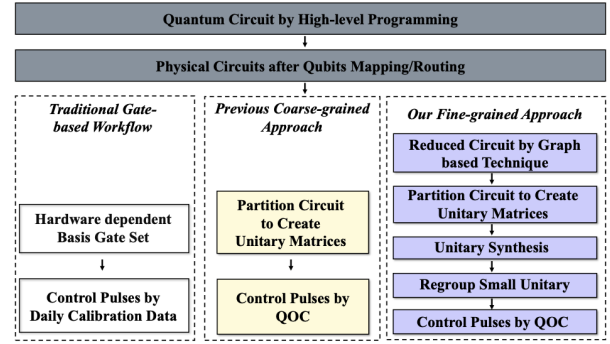


Fig. 1: The diagram compares the traditional gate-based workflow for creating quantum circuits with our novel fine-grained approach.

group. QOC is then applied to these unitary matrices to generate the optimized control pulses. While this approach can provide some benefits in terms of pulse optimization, it is limited by the inherent gate structure of the quantum circuit. If the depth of the circuit blocks does not match, leading to mismatched pulse latencies, the utilization rate of the qubit lines, where the pulses are applied, may not be optimal. This can result in suboptimal performance and longer overall execution times.

In this paper, we aim to propose a fine-grained approach in contrast to the previous coarse-grained methods and traditional workflow, as depicted in Figure 1. Thus, we introduce EPOC, a novel approach that combines ZX-calculus [20], circuit partitioning, and circuit synthesis with QOC to enhance pulse generation. Our main contribution is the development of a finer-grained method for generating quantum pulses, which results in increased parallelism and reduced latency. This is accomplished by partitioning circuits into blocks, computing their corresponding unitary matrices, and applying synthesis techniques to identify equivalent circuits composed of unitary gates. By breaking down relatively large circuit blocks into smaller unitary gates, we create additional opportunities for optimization, enabling more efficient and effective pulse generation. However, directly applying QOC to these small unitary gates, which are the products of synthesis, presents a challenge. The issue arises because these blocks are typically too small to fully leverage the advantages of QOC. To address this, we need to aggregate these blocks to achieve a suitable size that allows for effective application of QOC techniques. Therefore, we introduce a regrouping process for the unitary gates, where different unitary gates and two-qubit gates like CNOT are aggregated into a single unitary matrix. This regrouping technique allows us to further trade classical computational power for quantum computational power while limiting the size of the resulting matrices to avoid excessive computational overhead [18].

Finally, we apply QOC to the regrouped unitary matrices to obtain the optimized microwave pulses for the qubits. By combining QOC

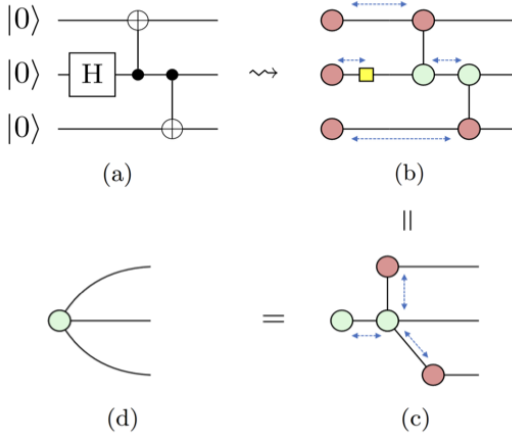


Fig. 2: ZX-Diagram of a GHZ-state preparation circuit. (a) Quantum circuit for preparing a 3-qubit GHZ state  $(|000\rangle + |111\rangle)/\sqrt{2}$ . (b) Corresponding ZX-diagram. Simplifications are made using rewriting rules such as spider fusion, which is indicated by dashed double-arrow lines for contractable spiders. (c) Further simplification. (d) Final compact representation of the GHZ state [21].

with the synthesized and regrouped unitary matrices, we generate highly optimized and robust quantum circuits tailored to the specific characteristics of the target quantum hardware.

The main contributions can be summarized as follows:

- We propose EPOC, a framework that identifies the optimal circuit size for combining QOC with synthesis techniques to enable efficient pulse generation while managing compilation overhead.
- We introduce a regrouping strategy that combines quantum gates into optimally-sized (not smallest but balanced) unitary matrices to achieve a balance between synthesis optimization potential and QOC benefits.
- We demonstrate through extensive benchmarking that this balanced approach achieves significant improvements in circuit latency and compilation efficiency compared to existing methods.

The remainder of this paper is organized as follows. Section I provides an introduction, followed by the background information in Section II. We introduce our proposed framework, EPOC, in Section III. The evaluation of EPOC is presented in Section IV. Section V discusses related work, and we conclude the paper in Section VI.

## II. BACKGROUND

### A. ZX-Calculus

The ZX-calculus, developed by Bob Coecke and Ross Duncan [22], [23], offers a graphical language tailored for representing quantum states and operations on qubits. It utilizes the formalism of string diagrams, with Pauli-X and Pauli-Z matrices as representing basis, thus termed ZX-diagrams. ZX-diagrams represent the topological configuration of quantum circuits, allowing transformations that do not alter qubit connections due to their topological invariance. The calculus includes a set of graphical rewrite rules, collectively known as the ZX-calculus rules. Fundamentally, ZX-diagrams are composed of building blocks called “spiders”, which represent various quantum computing elements such as isometries, states, unitary operations, and measurements in specific bases. These diagrams feature nodes colored in green and red, where green nodes correspond to the computational

basis states  $|0\rangle$  and  $|1\rangle$ , and red nodes represent the Hadamard-transformed basis states  $|+\rangle$  and  $|-\rangle$ . Additionally, Hadamard nodes are depicted as yellow boxes. Connections in the diagram are made by wires, which can curve, cross, and contract. In Fig. 2, we give an example showing the ZX-diagram of a GHZ state which illustrates the ZX-Calculus and the rewriting rule for circuit simplification.

The theoretical foundation of ZX-Calculus is rooted in the field of dagger compact categories [24], a discussion of which is beyond the scope of this paper but can be explored in further readings [25]. ZX-calculus has found application in various domains of quantum information and computation, including measurement-based quantum computation [26], quantum error correction [27], [28], and quantum circuit optimization [29]. The application of ZX-calculus to quantum circuit optimization is particularly relevant to ours.

### B. Synthesis

Quantum circuit synthesis plays a crucial role in quantum computing by generating efficient and optimized quantum circuits from high-level mathematical descriptions of quantum algorithms. Synthesis techniques aim to decompose unitary matrices, representing quantum transformations, into a sequence of elementary quantum gates that can be directly executed on quantum hardware [30]. The quality of synthesized circuits is typically evaluated based on the number of gates, particularly the count of expensive two-qubit gates like CNOT, as well as the overall circuit depth [31]. Shorter circuits with fewer gates are highly desirable, especially for NISQ devices characterized by limited coherence times and noisy operations. Effective synthesis methods can significantly reduce the computational overhead and improve the reliability of quantum algorithms on real hardware [32], [33]. Moreover, quantum circuit synthesis enables hardware design exploration, algorithm discovery, and circuit optimization. Existing synthesis approaches can be broadly categorized into top-down methods, which employ rule-based decomposition techniques, and bottom-up methods, which utilize numerical optimization and search strategies to construct circuits incrementally.

### C. Quantum Optimal Control

In the context of a closed quantum system, the dynamics can be described by a Hamiltonian, given as:

$$H(t) = H_0 + \sum_{j=1}^n u_j(t) H_j, \quad (1)$$

where  $H_0$  represents the drift Hamiltonian that accounts for the intrinsic evolution of the system, and  $H_j$  are the control Hamiltonians modulated by time-dependent control signals  $u_j(t)$ . The temporal evolution of the quantum state  $|\psi\rangle$  in this system is governed by the Schrödinger equation:

$$\frac{d}{dt} |\psi(t)\rangle = -iH(t)|\psi(t)\rangle, \quad (2)$$

assuming the initial state of the system at time  $t = 0$  is  $|\psi(0)\rangle$ .

QOC aims to determine the optimal control signals  $u_j(t)$  that transition the system from an initial quantum state to a desired target state. This control strategy is particularly critical in achieving precise state transformations while considering system constraints and minimizing operational errors. To optimize the control signals, QOC employs a cost function, typically defined in terms of the fidelity between the achieved unitary transformation induced by the applied control signals and a desired unitary goal. This fidelity metric quantifies the accuracy of the state transformation. Advanced algorithms such as Gradient Ascent Pulse Engineering (GRAPE) [34] and Chopped

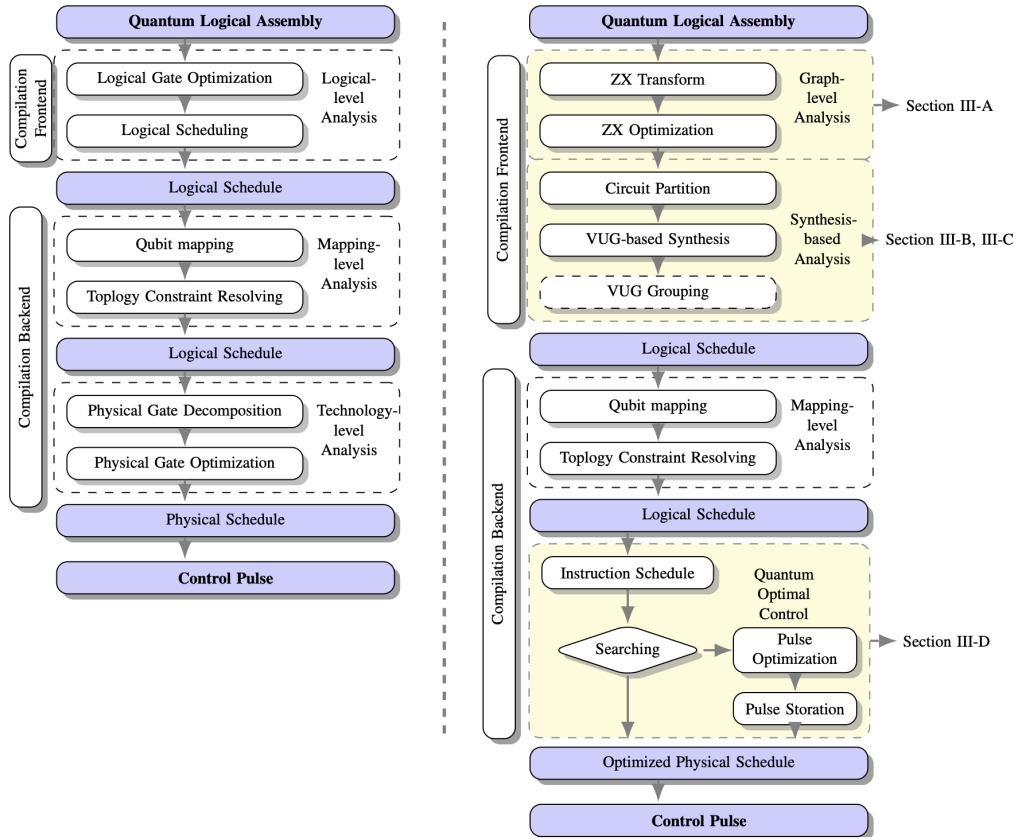


Fig. 3: The comparison between standard gate-based compilation (left) and our approach (right). The colored areas highlight the differences.

Random Basis (CRAB) [35] are utilized to systematically solve the QOC problem. These methods iteratively adjust the control signals to maximize the fidelity, ensuring that the final quantum state closely approximates the target state.

### III. METHODOLOGY

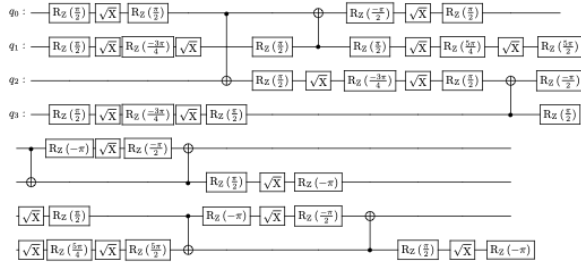
In this section, we present the methodologies employed in EPOC to accelerate pulse generation. As shown in Figure 3, EPOC consists of several steps, each of which will be discussed in detail throughout this section. We will provide a comprehensive explanation of the techniques and algorithms used in each stage of the EPOC process, highlighting their contributions to improving the efficiency and effectiveness of pulse generation.

#### A. Graph-based Depth Optimization

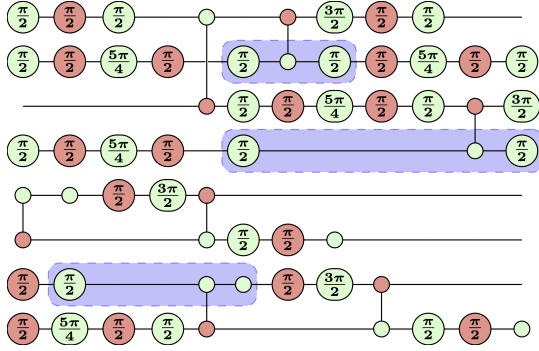
After converting the input circuit into a ZX graph, the gates in the circuit are analyzed to determine if they can be delayed and potentially cancelled against future gates or combined with other gates. This process, known as gate commutation, allows for circuit optimization and depth reduction. For instance, a NOT gate can be placed on either side of a CZ gate for both control and target qubits without altering the gate's functionality. Similarly, a Z gate on the control qubit can be commuted through a CNOT gate if a Z gate also appears on the target qubit, and a NOT gate on the target qubit can be commuted through a CNOT gate if a NOT gate also appears on the control qubit. However, a Hadamard gate cannot be commuted through a CZ gate or a CNOT gate due to its non-commutative properties. Gate commutation enables the aggregation of gates, leading to a reduction in circuit depth. For example, a NOT gate and a Hadamard gate can be combined into a Z gate, while a Z phase gate can be aggregated with a NOT gate, Z phase gate, or T gate by adjusting the rotation phase. Additionally, a pattern consisting

of a Z phase gate between two Hadamard gates is transformed into a Hadamard gate between two Z phase gates with negative phases to enhance commutativity and aggregation capabilities. Such a process is simplified with ZX graphs, as in a ZX graph, nodes in the same color are commutative and can be combined into a single node. Apart from gate commutation and gate aggregation, other ZX-calculus rules such as spider fusion are applied to simplify the circuit representations and further reduce the complexity of the quantum circuit. In our work, we utilize PyZX [36], a Python library designed for simplifying ZX-diagrams and optimizing quantum circuits.

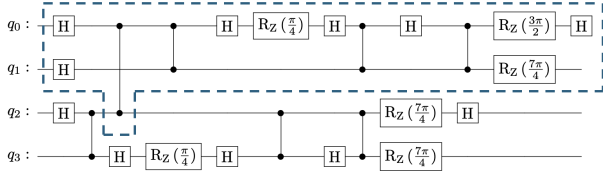
Figure 4 demonstrates a concrete example of the transformation and optimization process. In this example, we consider a bell state preparation circuit with four qubits. The first step involves transforming the original circuit into a ZX graph representation, under the rules provided by ZX-calculus. Once the quantum circuit has been represented as a ZX graph, we can employ optimization methods to simplify the circuit. As shown in Figure 4b, the nodes in the blue boxes of the same color are commutative and can be aggregated into a single node. We iteratively search for such patterns in the graph, ultimately resulting in a simplified and shallower quantum circuit. The optimized ZX graph can then be transformed back into a quantum circuit representation, and the circuit depth is reduced from 23 to 14, resulting in a more efficient and compact version of the original bell state preparation circuit. This example highlights the power of using ZX-calculus and graph-based optimization techniques to simplify quantum circuits. Furthermore, we remark that ZX-calculus for optimization is generally universal for most quantum circuits. For 34 randomly selected circuits, an average depth reduction of 1.48 times is observed as indicated in Figure 5. On extreme occasions, the circuit depth of a variational quantum eigensolver (VQE) is reduced from 7656 to 1110.



(a) Quantum Bell State Generation Circuit with 4 qubits. The circuit is implemented using the rotation gate, SX gate, and CNOT gate.



(b) ZX Graph Representation of the quantum circuit. The nodes within the blue box that share the same color are commutative.



(c) Example of circuit grouping for the optimized circuit after applying ZX-calculus optimization.

Fig. 4: Graph-level optimization steps of the Bell state preparation quantum circuit.

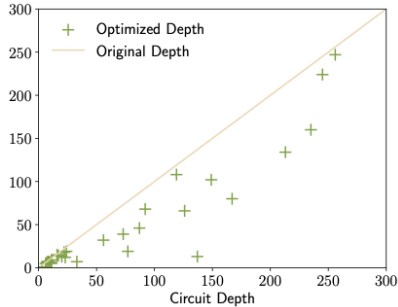


Fig. 5: ZX optimization results for 34 randomly selected quantum circuits. An average depth reduction of 1.48 times is observed across the selected circuits.

### B. Greedy Circuit Partition

After processing the circuit using ZX-calculus, we employ a greedy partitioning algorithm to divide it into smaller blocks for synthesis. The algorithm begins with horizontal cutting, grouping qubits together to form initial blocks, followed by vertical cutting to fill each block with as many gates as possible up to a predetermined size limit. By filling the blocks with gates, we ensure each block contains enough operations to benefit from synthesis, with minimal overhead

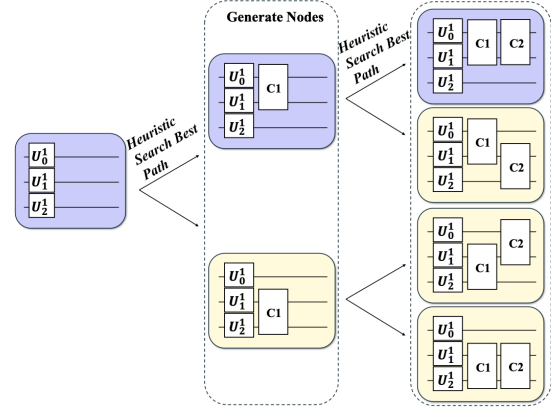


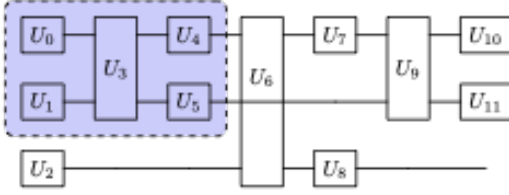
Fig. 6: Illustration of the quantum circuit synthesis process. The synthesis begins with an empty circuit and iteratively generates new nodes by inserting gates at different positions. Each node represents a candidate quantum circuit. The search for the optimal circuit follows the A algorithm, which evaluates nodes based on a cost function that measures the distance between the target unitary matrix and the unitary matrix of the synthesized circuit. The search continues until a circuit that accurately approximates the target unitary matrix with minimal size is found. The resulting synthesized circuit consists of variable unitary gates (VUGs) and CNOT gates.

introduced by vertical cutting. The prior ZX-calculus optimization enhances the effectiveness of the greedy partitioning algorithm. ZX-calculus simplifies the circuit by reducing it to a small gate set, resulting in a more uniform structure. This uniformity makes the circuit particularly suitable for the greedy partitioning approach, allowing for a balanced and efficient distribution of gates among the blocks. Figure 4c illustrates the partitioning process, showing a typical circuit block enclosed within a rectangular region after partitioning. These blocks are now ready for the subsequent circuit synthesis process, which will further optimize and transform them into a more compact representation. By combining ZX-calculus optimization with greedy circuit partitioning, we effectively decompose large quantum circuits into smaller, more manageable blocks.

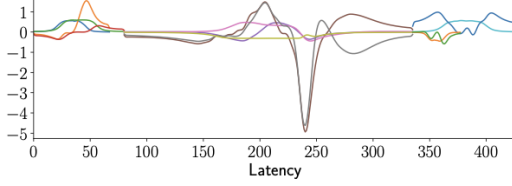
### C. Circuit Size Optimization: Balancing Synthesis and QOC

After the greedy circuit partitioning, we obtain a list of grouped gates. We then calculate the unitary matrices of these grouped gates. Since we limit the size of the groups, the computational overhead here is negligible and totally acceptable. In fact, we note that the overhead to compute the unitary matrices remains manageable even with very deep circuit blocks, as long as the number of qubits is moderate. Once we have the unitary matrices of circuit blocks, we leverage insights from QSearch [37] to develop our VUG-based circuit synthesis approach within the BQSKIT [38]. Our synthesis process generates a quantum circuit composed of VUGs and CNOT gates, as illustrated in Figure 6. The synthesized circuit shown in Figure 7a achieves a depth reduction from 11 to 7 compared to the original circuit in Figure 4c. However, a key observation emerges: while traditional synthesis approaches aim to minimize circuit size, this may not be optimal when combining synthesis with QOC. Our analysis reveals that fine-grained VUGs limit the potential benefits of QOC. This insight led us to develop a strategic regrouping approach that creates optimally-sized unitary matrices for QOC processing. Figures 7b and 7c demonstrate the significant impact of this approach: direct pulse generation from individual VUGs results in higher circuit

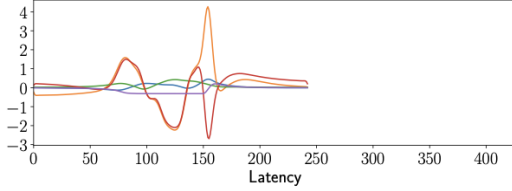




(a) The synthesis result of the partitioned circuit in the blue box of Fig 4c.



(b) Illustration of the Pulse generation for each gate within the purple box in Fig 7a.



(c) Pulse generation for the entire purple box in Fig 7a, treating all gates as a single grouped unitary.

Fig. 7: The comparison between circuit synthesis + QOC before regrouping strategy and after implement regrouping strategy.

latency, while our regrouping strategy (purple box in Figure 7c) substantially reduces pulse latency. This finding emphasizes the importance of identifying the sweet spot in circuit size that maximizes the combined benefits of synthesis and QOC.

#### D. Pulse Generation

In our proposed framework EPOC, our methodologies are built upon prior works AccQOC [18] and PAQOC [19]. We employ QOC techniques to generate a pulse library, which essentially serves as a lookup table. The indices of the table are unitary matrices, and each entry stores the corresponding microwave pulses. The techniques in AccQOC and PAQOC involve a binary search to determine a short pulse latency for a given unitary matrix as input. Specifically, the QOC we employ is implemented using the GRAPE algorithm [34], where pulses are divided into segments or time slots with constant values, forming a pulse sequence. The GRAPE method optimizes the time slots according to a cost function, often defined as the fidelity between the pulse unitary and the target unitary, adjusting pulse strengths during each time slot based on gradients. However, pulse simulation and optimization are computationally expensive, which limits the scalability of QOC; thus, we limit the size of the unitary matrices input to the QOC algorithm. Once we have obtained the pulse library, we can accelerate future QOC processes by directly fetching pre-existing pulses for unitary matrices already present in the library. During this process, we also consider the global phase when comparing unitary matrices. In contrast to the previous works on accelerating QOC, EPOC supports the detection of unitary similarity with global phase, allowing us to identify more matched unitary matrices, similar to having a higher cache hit rate.

#### IV. EVALUATION

The experiments were conducted on a Linux-based system with Debian 5.10.179-1 operating system, using 8 nodes. Each node is equipped with 32 cores and 256GB of RAM.

As the first to introduce synthesis into the workflow of QOC, we compare the results from: (1) applying QOC directly on the synthesized circuits and (2) grouping the synthesized results before running QOC. As shown in Figure 8, Figure 9 and Figure 10, we demonstrate although both settings have similar compilation times, our method with an extra grouping step significantly reduces circuit latency and increases circuit fidelity. We evaluate our methods using 17 benchmarks from QASMBench [39], showcasing the effectiveness of incorporating the grouping step in the QOC workflow.

The fidelity is calculated as the accumulated fidelity of each quantum pulse. Specifically, we calculate the fidelity of each pulse and then multiply these fidelities together for all the pulses in the circuit. The fidelity of each individual pulse is determined by the distance between the unitary matrices before and after QOC. This calculation can be described by the following equation:

$$ESP = \prod_{i=1 \dots n} (1 - |U_i - H_i(t)|) \quad (3)$$

where  $ESP$  represents the estimated total fidelity of the quantum circuit,  $n$  is the number of pulses,  $U_i$  is the target unitary matrix for the  $i$ -th pulse, and  $H_i(t)$  is the unitary matrix achieved by the optimized pulse after applying QOC. The term  $|U_i - H_i(t)|$  calculates the distance between the target and optimized unitary matrices for each pulse, and  $(1 - |U_i - H_i(t)|)$  represents the fidelity of the  $i$ -th pulse. The product of these pulse fidelities gives the overall fidelity of the entire quantum circuit.

Across the 17 benchmarks from QASMBench [39], our method achieves an average reduction of 51.11% in pulse latency with only a 7.11% increase in compilation time by incorporating a regrouping step after synthesis in the QOC workflow. Furthermore, our approach manages to increase the circuit fidelity by an average of 33.77%. These results demonstrate that the regrouping step is necessary and beneficial for reducing circuit latency without introducing significant overhead, highlighting the effectiveness of our proposed method in enhancing the performance of quantum circuits.

We compare our results with both PAQOC [19] and AccQOC [18], two recent works on quantum optimal control. As shown in Table I, our approach achieves significant latency reductions: 31.74% compared to PAQOC and 71.23% compared to AccQOC. Moreover, we consistently report higher fidelity values compared to both approaches - our method achieves an average fidelity of 0.974, surpassing AccQOC's average of 0.890 across all circuits, while also outperforming PAQOC's reported fidelities where data is available. This dual improvement in both latency and fidelity demonstrates the effectiveness of our framework. We attribute these improvements to our novel circuit synthesis process, which effectively optimizes quantum circuits before the QOC step, resulting in reduced latency and increased fidelity. While our synthesis process adds extra computational workload compared to both PAQOC and AccQOC, it mainly involves local entanglement and unitary calculations that can be executed in parallel. Therefore, our approach maintains scalability while significantly reducing pulse latency. We validated our framework by testing it with a large and deep 160-qubit quantum program, obtaining meaningful results. Although this data point isn't in our table due to the lack of comparable data from other methods, it confirms the feasibility and scalability of our approach.

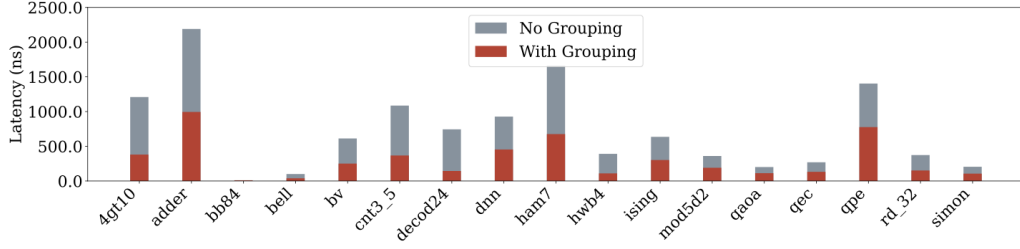


Fig. 8: The latency before and after grouping. We show through the results that the grouping can reduce the circuit latency.

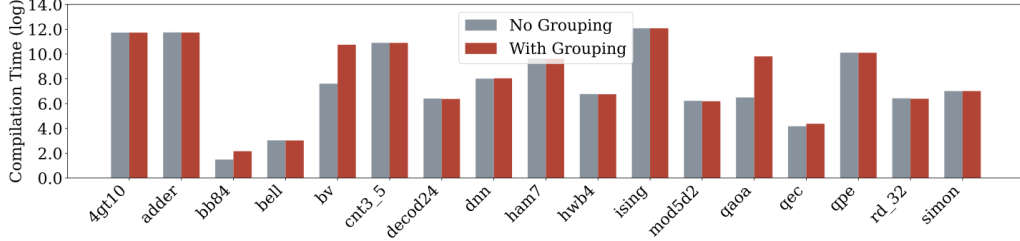


Fig. 9: The compilation time with and without grouping of quantum gates before the final QOC step. This indicates that the latency reduction benefits can be achieved through grouping without incurring a significant increase in compilation time.

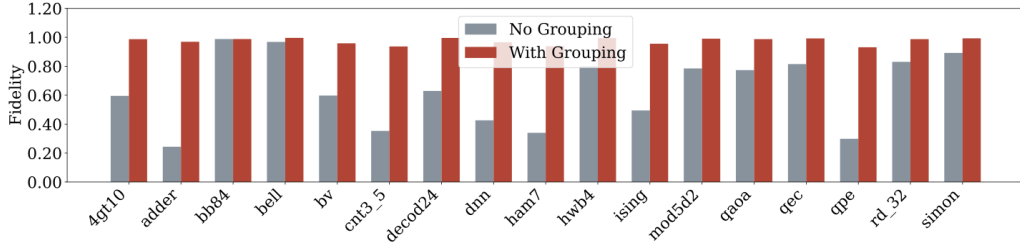


Fig. 10: The fidelity of quantum circuits with and without grouping. The results can be attributed to the fact that the no-grouping approach operates at a very fine granularity, which can lead to the accumulation of errors during the QOC process.

| Circuit | Gate-based | Latency (ns) |            |              | AccQOC [18] | Fidelity   |              |
|---------|------------|--------------|------------|--------------|-------------|------------|--------------|
|         |            | AccQOC [18]  | PAQOC [19] | EPOC         |             | PAQOC [19] | EPOC         |
| simon   | 469        | 175.22       | 141.23     | <b>92</b>    | 0.91        | -          | <b>0.984</b> |
| bb84    | 56.5       | 24.07        | 13         | <b>10</b>    | 0.659       | 0.981      | <b>0.988</b> |
| bv      | 901        | 345.16       | 321        | <b>268.5</b> | 0.941       | 0.971      | 0.968        |
| qaoa    | 1324.5     | 370.75       | 393        | <b>111.5</b> | 0.903       | 0.952      | <b>0.984</b> |
| decod24 | 1315.5     | 382.58       | 315        | <b>144</b>   | 0.953       | 0.982      | <b>0.989</b> |
| dnn     | 3174.5     | 606.67       | 385        | <b>453.5</b> | 0.941       | -          | <b>0.965</b> |
| ham7    | 5238.5     | 1553.09      | 1186.5     | <b>675.5</b> | 0.920       | -          | <b>0.938</b> |

TABLE I: Comparison between AccQOC [18], PAQOC [19], and EPOC (our approach) for 7 widely applied quantum circuits. In most scenarios, EPOC returns a generation result with shortened latency and increased fidelity. On average, our approach achieves a 31.74% reduction in latency compared to PAQOC [19] and a 76.80% reduction compared with the gate-based method to create pulse sequences. The ‘-’ indicates PAQOC returns fidelity as 0 for the corresponding task.

## V. RELATED WORK

A recent work AccQOC [18] employs a novel approach by segmenting the quantum circuit into small, uniform subcircuits, each comprising two qubits. It then builds a pulse database for these subcircuits, storing previously generated pulses for efficient reuse. To streamline this process, AccQOC utilizes a similarity graph that maps the distances between different fixed-size subcircuits, including those not currently in the database. The Minimum Spanning Tree (MST) of this graph is then calculated to optimize the order in which control pulses are constructed for the subcircuits. This method significantly enhances the efficiency of pulse compilation, reducing overall time requirements. A more recent work, PAQOC [19], is automatically de-

tecting common gate patterns through subgraph mining for expanded search space exploration. It also systemically constructs customized gate-sets based on their impact on overall program latency and quickly adapts to system recalibrations with its small-scale pattern-based gate generation. However, all the aforementioned compilation strategies generate pulses directly from the partitioned circuit group. Therefore, the group size is constrained due to the unmanageable overhead of pulse generation for large unitaries.

## VI. CONCLUSION

In this paper, we introduced the EPOC framework, integrating ZX-calculus, circuit partitioning, and synthesis techniques with QOC to enhance pulse generation efficiency in quantum circuits. EPOC employs refined granular optimization to significantly reduce circuit latency and computational overhead while increasing parallelism. By strategically regrouping unitary matrices and optimizing their configurations, EPOC improves the scalability of QOC processes and demonstrates considerable performance enhancements in NISQ devices. We achieve notable improvements in pulse generation speed and circuit performance, including an average 31.74% reduction in circuit latency over PAQOC and 71.23% compared to AccQOC. These results indicate significant potential for applying EPOC in practical quantum computing scenarios.

## VII. ACKNOWLEDGMENT

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